

Executive Engineering Report

Next-Generation Autonomous Pipeline Inspection Infrastructure

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Executive Summary

This report outlines the technical milestones, efficiency gains, and deployment roadmap for the automated robotics inspection stack. Deployment across key test facilities demonstrated a **72% reduction** in manual inspection hours alongside a **99.4% spatial tracking fidelity**.

1 Strategic Key Performance Indicators (KPIs)

Metric	Baseline	Target	Business Impact
Inspection Turnaround	4.5 hrs	1.2 hrs	Accelerated operational readiness.
Spatial Resolution	±5.0 cm	±0.5 cm	Precise defect localization for compliance.
Sensor Sampling Rate	10 Hz	100 Hz	High-density environmental telemetry.

2 Key Technology Highlights

- **High-Frequency Micro-Odometry:** Ingests wheel encoder and 9-DoF IMU telemetry at 100 Hz with zero-drift deadband filtering.
- **Synchronized DTW Video Localization:** Automatically matches before-and-after duct cleaning videos using distance-indexed keyframing.